MULTICOLLINEARITY

Q1) What does it mean that the model has "Multicollinearity"?

- The column vectors of the design matrix are almost linearly independent

- In the multiple linear regression model $y = X\beta + \epsilon$,
  $\exists \beta_j \in \beta$ s.t. $\beta_j \simeq 0$

- Suppose that there is a high collinearity between x_i and x_j $(i \neq j)$
  $\exists a \in \mathbb{R}^p$ s.t. $a^T X_{(i)} \simeq 0$ (a is a non-zero vector)

Q2) Prove Theorem 2

Coefficient of determination of the MLRM for the j^{th} predictor for $1 \le j \le p$ is :

$$R_j^2 = 1 - \frac{x_{j,\perp}^T x_{j,\perp}}{S_{jj}}$$

response/dependent variable

predictor/independent variable

$$x_j = x_{(j)} a + \delta$$

$$x_{(j)} = (\mathbb{1}, x_1, \cdots, x_{j-1}, x_{j+1}, \cdots, x_p)$$

$$x_{j,\perp}^T = x_j - \Pi_{x_{(j)}} x_j$$

$$S_{jj} = \sum_{j=1}^{n} (x_{ij} - \bar{x}_j)^2$$

(Approach #1)

$y = x_j$

$X = x_{(j)}$

$$R^2 = \frac{SSR}{SST} = \frac{y^T(\Pi_X - \Pi_{\mathbb{1}})y}{y^T(I - \Pi_{\mathbb{1}})y} = \frac{x_j^T(\Pi_{X_{(j)}} - \Pi_{\mathbb{1}})x_j}{x_j^T(I - \Pi_{\mathbb{1}})x_j}$$

$$= \frac{x_j^T(\Pi_{X_{(j)}} - I + I - \Pi_{\mathbb{1}})x_j}{x_j^T(I - \Pi_{\mathbb{1}})x_j}$$

$$= 1 - \frac{x_j^T(I - \Pi_{x_{(j)}})x_j}{x_j^T(I - \Pi_{\mathbb{1}})x_j}$$

$$= 1 - \frac{\| x_j - \Pi_{x_{(j)}} x_j \|^2}{\sum_{i=1}^{n} (x_{ij} - \bar{x}_i)^2} = 1 - \frac{x_{j,\perp}^T x_{j,\perp}}{S_{jj}}$$

(Approach #2)

$$R_j^2 = \frac{SSR_j}{SST_j} = 1 - \frac{SSE_j}{SST_j}$$

$$SSE_j = \| y - \hat{y} \|^2 = \| X_j - X_{(j)}\hat{a} \|^2$$
$$= \| X_j - X_{(j)}(X_{(j)}^T X_{(j)})^{-1} X_{(j)}^T X_j \|^2$$
$$= \| X_j - \Pi_{X_{(j)}} X_j \|^2$$
$$= X_{j,\perp}^T X_{j,\perp}$$

$$SST_j = \sum_{i=1}^{n} (y_i - \bar{y})^2 = \sum_{i=1}^{n} (x_{ij} - \bar{x}_j)^2 = S_{jj}$$

$$\therefore R_j^2 = 1 - \frac{X_{j,\perp}^T X_{j,\perp}}{S_{jj}}$$

Q3) Prove Theorem 3

$$y = X\beta + \epsilon, \quad \varepsilon_1, \cdots, \varepsilon_n \overset{i.i.d}{\sim} (0, \sigma^2)$$

$$Var(\hat{\beta}_j) = \frac{1}{1 - R_j^2} \cdot \frac{\sigma^2}{S_{jj}}, \quad S_{jj} = \sum_{i=1}^{n} (x_{ij} - \bar{x}_j)^2$$

(Approach #1)

$$Var(\hat{\beta}_j) = Var(e_j^T \hat{\beta})$$
$$= e_j^T Var(\hat{\beta}) e_j$$
$$= e_j^T \sigma^2 (X^T X)^{-1} e_j$$

$$(X^T X) = (X_j \ X_{(j)})^T (X_j \ X_{(j)})$$
$$= \begin{bmatrix} X_j^T X_j & X_j^T X_{(j)} \\ X_{(j)}^T X_j & X_{(j)}^T X_{(j)} \end{bmatrix}$$

$$(X^T X)^{-1} = \begin{bmatrix} A_{11\cdot 2}^{-1} & \cdots \\ \cdots & \cdots \end{bmatrix}$$

$$A_{11\cdot 2} = A_{11} - A_{12} A_{22}^{-1} A_{21}$$
$$= X_j^T X_j - X_j^T X_{(j)} (X_{(j)}^T X_{(j)})^{-1} X_{(j)}^T X_j$$
$$= X_j^T (I - \Pi_{X_{(j)}}) X_j$$
$$= X_{j,\perp}^T X_{j,\perp}$$

$$e_j^T (X^T X)^{-1} e_j = X_{j,\perp}^T X_{j,\perp}$$

$$\therefore Var(\hat{\beta}_j) = \sigma^2 (X_{j,\perp}^T X_{j,\perp})^{-1}$$

$$R_j^2 = 1 - \frac{X_{j,\perp}^T X_{j,\perp}}{S_{jj}}$$

$$\therefore X_{j,\perp}^T X_{j,\perp} = (1 - R_j^2) \cdot S_{jj}$$

$$\Rightarrow Var(\hat{\beta}_j) = \sigma^2 (X_{j,\perp}^T X_{j,\perp})^{-1} = \frac{1}{1 - R_j^2} \cdot \frac{\sigma^2}{S_{jj}}$$

(Approach #2)

$$\hat{\beta}_j = (X_{j,\perp}^T X_{j,\perp})^{-1} X_{j,\perp}^T y, \quad Var(\hat{\beta}_j) = \sigma^2 (X_{j,\perp}^T X_{j,\perp})^{-1}$$

We know that in a MLRM: $y = X_1\gamma_1 + X_2\gamma_2 + \epsilon$

$$\hat{\gamma}_2 = (X_{2,\perp}^T X_{2,\perp})^{-1} X_{2,\perp}^T y \quad \text{and} \quad Var(\hat{\gamma}_2) = \sigma^2 (X_{2,\perp}^T X_{2,\perp})^{-1}, \quad X_{2,\perp} = X_2 - \Pi_{X_1} X_2$$

Let $X_1 = X_{(j)}$, $X_2 = X_j$, $\gamma_2 = \beta_j$

$$X_{2,\perp} = (I - \Pi_{X_{(j)}}) X_j = X_{j,\perp}$$

From Theorem 2, $R_j^2 = 1 - \dfrac{x_{j,\perp}^T x_{j,\perp}}{S_{jj}}$

$$\therefore \frac{1}{1 - R_j^2} = \frac{S_{jj}}{x_{j,\perp}^T x_{j,\perp}}$$

$$\therefore Var(\hat{\beta}_j) = \frac{1}{1 - R_j} \cdot \frac{\sigma^2}{S_{jj}}$$

* note : x_j 라는 변수가 나머지 변수들 $x_{(j)}$에 의해서 잘 설명이 된다면 (i.e., $R^2 \approx 1$) $\hat{\beta}_j$의 추정 정확도/신뢰도가 굉장히 낮아지게 된다.

Q4) What is the "Variation Inflation Factor (VIF)"?

$$VIF_j : \frac{1}{1-R_j^2} \quad (j = 1, \cdots, p)$$

: variation inflation factor.

Q5) Prove the following:

Consider the MLRM: $y = X\beta + \epsilon, \ \epsilon_1, \cdots, \epsilon_n \overset{i.i.d}{\sim} (0, \sigma^2)$

Decompose X and β as: $X = (\mathbb{1} \ X_1), \ \beta = (\beta_0 \ \gamma^T)^T$

Rewrite the regression model as:

$$y = \mathbb{1}\beta_0 + X_1\gamma + \epsilon = \mathbb{1}\beta_0' + X_c\gamma + \epsilon$$
$$X_c = (I_n - \Pi_{\mathbb{1}})X_1$$
$$\beta_0' = \beta_0 + n^{-1}\mathbb{1}^T X_1 \gamma$$

$$\begin{aligned} y = \mathbb{1}\beta_0 + X_1\gamma_1 &= \mathbb{1}\beta_0 + \Pi_{\mathbb{1}}X_1\gamma_1 + (X_1 - \Pi_{\mathbb{1}}X_1)\gamma_1 \\ &= \mathbb{1}\beta_0 + \Pi_{\mathbb{1}}X_1\gamma_1 + X_{1,\perp}\gamma_1 \\ &= \mathbb{1}\beta_0 + \mathbb{1}(\mathbb{1}^T\mathbb{1})^{-1}\mathbb{1}^T\gamma_1 + X_{1,\perp}\gamma_1 \\ &= \mathbb{1}(\beta_0 + \tfrac{1}{n}\mathbb{1}^T\gamma_1) + X_{1,\perp}\gamma_1 \\ &= \mathbb{1}\beta_0' + X_c\gamma_1 \end{aligned}$$

Q6) Prove that the MLRM suffers from multicollinearity if and only if the centered version does as well.

In general MLRM, multicollinearity means:

$$a_0\mathbb{1} + a_1x_1 + \cdots + a_px_p \approx 0$$

In centered MLRM, multicollinearity means:

$$b_1(x_1 - \bar{x}_1\mathbb{1}) + \cdots + b_p(x_p - \bar{x}_p\mathbb{1}) \approx 0$$

(centered $\rightarrow$ general)

$$\begin{aligned} & b_1(x_1 - \bar{x}_1\mathbb{1}) + \cdots + b_p(x_p - \bar{x}_p\mathbb{1}) \\ &= b_1x_1 + \cdots + b_px_p - (b_1\bar{x}_1 + \cdots + b_p\bar{x}_p)\mathbb{1} \approx 0 \end{aligned}$$

(general $\rightarrow$ centered)

$$a_0\mathbb{1} + a_1x_1 + \cdots + a_px_p \approx 0 \quad \cdots ①$$
$$\Rightarrow \Pi_{\mathbb{1}}(a_0\mathbb{1} + a_1x_1 + \cdots a_px_p) \approx 0$$
$$\therefore a_0\mathbb{1} + a_1\bar{x}_1 + \cdots a_p\bar{x}_p \approx 0 \quad \cdots ②$$

$$① - ② = a_1(x_1 - \bar{x}_1) + \cdots + a_p(x_p - \bar{x}_p) \approx 0$$

Q7) Show that the $\hat{\beta}_0', \hat{\gamma}$ estimates from $y = \mathbb{1}\beta_0' + X_c\gamma + \epsilon$ and $\hat{\beta}_0', \hat{\gamma}$ estimates from $y = \mathbb{1}\beta_0' + \epsilon$ and $y_c = X_c\gamma + \epsilon$ respectively are equivalent.

$\hat{\beta}_0'$ and $\hat{\gamma}$ estimates from $y = \mathbb{1}\beta_0' + X_c\gamma + \epsilon$

$$\hat{\beta}_0' = (\mathbb{1}^T\mathbb{1})^{-1}\mathbb{1}^Ty, \quad \hat{\gamma} = (X_c^TX_c)^{-1}X_c^Ty$$

$\hat{\beta}_0'$ and $\hat{\gamma}$ estimates from $y = \mathbb{1}\beta_0' + \epsilon$ and $y_c = X_c\gamma + \epsilon$

$$\begin{aligned} \hat{\beta}_0' = (\mathbb{1}^T\mathbb{1})^{-1}\mathbb{1}^Ty, \quad \hat{\gamma} &= (X_c^TX_c)^{-1}X_c^Ty_c \\ &= (X_c^TX_c)^{-1}X_c^T(y - \Pi_{\mathbb{1}}y) \\ &= (X_c^TX_c)^{-1}X_c^Ty \quad \because X_c^T\Pi_{\mathbb{1}}y = 0 \end{aligned}$$

Q8) What is the Mathematical representation of a "Principal Component".

Consider the centered MLRM: $y_c = X_c\gamma + \epsilon$ where $\epsilon_1, \cdots, \epsilon_n \overset{iid}{\sim} (0, \sigma^2)$

Since $X_c^TX_c$ is symmetric, it can be decomposed into:

$$X_c^TX_c = P\Lambda P^T, \quad PP^T = P^TP = I, \quad \Lambda = diag(\lambda_1, \lambda_2, \cdots, \lambda_n)$$
$$P = [v_1 \cdots v_p]$$

v_j are orthonormal eigenvectors of $X_c^TX_c$ ordered in terms of the respective eigenvalues $\lambda_1 \geq \cdots \geq \lambda_p \geq 0$ ($\because X_c^TX_c$ is non-negative definite)

By letting $v_j = (v_{1j} \cdots v_{pj})^T$, the following vector

$$z_j = X_cv_j = \sum_{i=1}^{p} v_{ij}(x_i - \bar{x}_i\mathbb{1}) \quad (j = 1, \cdots, p)$$

is called a "Principal Component".

$$\begin{aligned} z_j = X_cv_j &= (x_1 - \bar{x}_1\mathbb{1}, \cdots, x_p - \bar{x}_p\mathbb{1})\begin{pmatrix} v_{1j} \\ \vdots \\ v_{pj} \end{pmatrix} \\ &= \sum_{i=1}^{p} v_{ij}(x_i - \bar{x}_i\mathbb{1}) \end{aligned}$$

Q9) Prove Theorem 6

$z_j = j^{th}$ principal component $(j = 1, \cdots, p)$
$\|z_j\|^2 = \lambda_j$ and $\langle z_j, z_i \rangle = 0 \ (j \neq i)$

$$P = [\,v_1 \cdots v_p\,]$$

$z_j = X_c v_j = X_c P e_j$

$$\begin{aligned} \|z_j\|^2 = z_j^T z_j &= e_j^T P^T X_c^T X_c P e_j \\ &= e_j^T P^T P \Lambda P^T P e_j \\ &= e_j^T \Lambda e_j \\ &= \lambda_j \end{aligned}$$

$$\begin{aligned} \langle z_j, z_i \rangle &= e_j^T P^T X_c^T X_c P e_i \\ &= e_j^T P^T P \Lambda P^T P e_i \\ &= e_j^T \Lambda e_i \\ &= 0 \end{aligned}$$

(TEXTBOOK APPROACH)

$X_c P = (z_1 \cdots z_p)$

$$(X_c P)^T X_c P = \begin{bmatrix} z_1^T \\ \vdots \\ z_p^T \end{bmatrix} \begin{bmatrix} z_1 \cdots z_p \end{bmatrix} = \begin{bmatrix} z_1^T z_1 & z_1^T z_2 & \cdots & z_1^T z_p \\ \vdots & & & \\ z_p^T z_1 & z_p^T z_2 & \cdots & z_p^T z_p \end{bmatrix} = \Lambda = diag(\lambda_1 \cdots \lambda_p)$$

$\therefore z_i^T z_i = \|z_i\|^2 = \lambda_i$ and $\langle z_j, z_i \rangle = 0$

Q10) Prove Theorem 7

Under the centered MLRM : $y = \mathbb{1}\beta_0' + X_c \gamma + \epsilon$

$$\sum_{j=1}^{p} Var(\hat{\beta}_j) = \sigma^2 \sum_{j=1}^{p} \frac{1}{\lambda_j}$$

$\hat{\beta}_j = e_j^T \hat{\beta} = e_j^T (X_c^T X_c)^{-1} X_c^T y$

$$\begin{aligned} \sum_{j=1}^{p} Var(\hat{\beta}_j) &= \sum_{j=1}^{p} e_j^T \sigma^2 (X_c^T X_c)^{-1} e_j = \sigma^2 \sum_{j=1}^{p} e_j^T (P \Lambda P^T)^{-1} e_j \\ &= \sigma^2 \sum_{j=1}^{p} e_j^T P \Lambda^{-1} P^T e_j \end{aligned}$$

Let $P = \begin{bmatrix} w_1 \\ \vdots \\ w_p \end{bmatrix}$, w_i are orthonormal rows.

$P\Lambda^{-1}P^T = diag(\lambda_i^{-1} w_i w_i^T) = diag(\lambda_i^{-1})$

$$\begin{aligned} \therefore \sigma^2 \sum_{j=1}^{p} e_j^T P \Lambda P^T e_j &= \sigma^2 \sum_{j=1}^{p} e_j^T diag(\lambda_1^{-1} \cdots \lambda_j^{-1} \cdots \lambda_p^{-1}) e_j \\ &= \sigma^2 \sum_{j=1}^{p} \frac{1}{\lambda_j} \end{aligned}$$

(APPROACH #2)

$$\begin{aligned} \sum_{j=1}^{n} Var(\hat{\beta}_j) &= trace(Var(\hat{\beta})) = trace(\sigma^2 (X_c^T X_c)^{-1}) \\ &= trace(\sigma^2 P \Lambda^{-1} P^T) = \sigma^2 trace(\Lambda^{-1} P^T P) \\ &= \sigma^2 trace(\Lambda^{-1}) \\ &= \sigma^2 \sum_{i=1}^{p} \frac{1}{\lambda_i} \end{aligned}$$

Q11) Prove that the following are equivalent :

(a) The j^{th} eigenvalue λ_j of $X_c^T X_c$ is zero.
(b) The j^{th} principal component z_j is also a zero vector
(c) The column vectors of the matrix X_c are linearly independent.

(a) $\Leftrightarrow$ (b)
: $\|z_j\|^2 = \lambda_i$ $\therefore$ if $\lambda_i = 0$, $\|z_i\|^2 = 0 \Leftrightarrow z_i = 0$

(b) $\Leftrightarrow$ (c)

$$z_j = \sum_{i=1}^{p} v_{ij} \underbrace{(x_i - \bar{x}_i \mathbb{1})}_{X_c\text{'s column vector}}$$

$P = [\,v_1 \cdots v_p\,] \in \mathbb{R}^{p \times 1}$ $\quad \because X_c^T X_c = P \Lambda P^T \in \mathbb{R}^{p \times p}$
$v_1 = [\,v_{11} \cdots v_{p1}\,]^T$

$$0 = z_j = \sum_{i=1}^{p} v_{ij} (x_i - \bar{x}_i \mathbb{1})$$

$\therefore X_c$'s column vectors are linearly independent. $\square$

RIDGE REGRESSION

Q12) Define the "Ridge Regression".

The "Ridge Regression" is one variant of penalized square estimation.

The "ridge estimator" is defined as a minimizer of

$$\hat{\gamma}^R(k) = \underset{\gamma \in \mathbb{R}^p}{\operatorname{argmin}} \left[\| y_c - X_c\gamma \|^2 + k\|\gamma\|^2 \right], \quad \text{for some } k \geq 0$$

Q13) Prove Theorem 9

$$\hat{\gamma}^R_{(k)} = (X_c^T X_c + kI_p)^{-1} X_c^T y_c$$

$$L(\gamma) = \| y_c - X_c\gamma \|^2 + k\|\gamma\|^2 = y_c^T y_c - 2y_c^T X_c \gamma + \gamma^T X_c^T X_c \gamma + k\gamma^T\gamma$$

$$\partial L / \partial \gamma = -2X_c^T y_c + 2(X_c^T X_c)\gamma + 2k\gamma = 0$$

$$2(X_c^T X_c)\gamma - 2k\gamma = 2X_c^T y_c$$

$$(X_c^T X_c - kI_p)\gamma = X_c^T y_c$$

$$\hat{\gamma}^R(k) = (X_c^T X_c - kI_p) X_c^T y_c$$

$\nabla^2 L(\gamma) = 2X_c^T X_c + 2kI_p$: positive definite matrix.

$$\because X_c^T X_c + kI_p = P\Lambda P^T + kPP^T$$

$$= P(\Lambda + k)P^T, \quad \Lambda = \operatorname{diag}(\lambda_1 + k, \cdots, \lambda_p + k)$$

$$\lambda_1 + k \geq \cdots \geq \lambda_p + k > 0$$

Since all the eigenvalues of $X_c^T X_c + kI_p$ are strictly positive, $P(\Lambda + k)P^T$ is a positive definite matrix.

* Note: Suppose that X_c is full rank and nearly linearly dependent. Then, for $\lambda_1 \geq \cdots \geq \lambda_p \geq 0$, the last few eigenvalues are very close to zero. In such case, the calculation of $(X_c^T X_c)^{-1}$ becomes overly complicated. Therefore, we add k before finding the inverse matrix. (Ex. $\frac{1}{x} \gg 0$ if $x \approx 0$)

Q14) Prove Theorem 10

Consider the centered MLRM

$$\underset{\gamma \in \mathbb{R},\, \|\gamma\|^2 \leq d}{\operatorname{argmin}} \| y_c - X_c\gamma \|^2 = \hat{\gamma}^R(k) \quad = \underbrace{\underset{\gamma \in \mathbb{R}^p :\, \|\gamma\|^2 \leq d}{\operatorname{argmin}} (\gamma - \hat{\gamma})^T X_c^T X_c (\gamma - \hat{\gamma}),}_{\text{2nd equation}}$$

$\hat{\gamma}$: OLS estimator

d : $\hat{\gamma}^T [I_p + k(X_c^T X_c)]\hat{\gamma}$

$$\hat{\gamma}^R(k) = \underset{\|\gamma\|^2 \leq d}{\operatorname{argmin}} \| y_c - X_c\gamma \|^2$$

$$d = \hat{\gamma}^T (I_p + k(X_c^T X_c)^{-1})^{-2} \hat{\gamma}$$

* derive $\hat{\gamma}^R(k)$ using the Lagrange multiplier method. &

* show that first and second equations are equivalent.

* $\underset{\|\gamma\|^2 \leq d}{\operatorname{argmin}} \| y_c - X_c\gamma \|^2 \Rightarrow \mathcal{U}(\gamma, \lambda) = \| y_c - X_c\gamma \|^2 + \lambda(\|\gamma\|^2 - d) \quad (\gamma \in \mathbb{R}^p, \lambda > 0)$

↳ 라그랑주 승수

Lagrange Multiplier

(1) For fixed $\lambda > 0$, find $\hat{\gamma}(\lambda) = \underset{\gamma}{\operatorname{argmin}}\, \mathcal{U}(\gamma, \lambda)$

(2) Find $\lambda^* > 0$ s.t. $\|\hat{\gamma}(\lambda^*)\|^2 = d$.

WTS: $\underbrace{\hat{\gamma}(\lambda^*)}_{*} = \underset{\|\gamma\| \leq d}{\operatorname{argmin}} \| y_c - X_c\gamma \|^2$

* : $\forall \gamma$: $\|\gamma\|^2 \leq d \Rightarrow \| y_c - X_c\gamma \|^2 \geq \| y_c - X_c\gamma \|^2 + \lambda^*(\|\gamma\|^2 - d)$

$$= \mathcal{U}(\gamma, \lambda^*)$$

$$\geq \mathcal{U}(\hat{\gamma}(\lambda^*), \lambda^*)$$

$$= \| y_c - X_c\hat{\gamma}(\lambda^*) \|^2 + \lambda(\|\hat{\gamma}(\lambda^*)\|^2 - d)$$

$$\therefore \| y_c - X_c\gamma \|^2 \geq \mathcal{U}(\hat{\gamma}(\lambda^*), \lambda^*)$$

(1) $\underset{\gamma}{\operatorname{argmin}}\, \mathcal{U}(\gamma, \lambda) = \underset{\gamma}{\operatorname{argmin}} (\| y_c - X_c\gamma_c \|^2 + \lambda\|\gamma\|^2 - \lambda d)$

$= \underset{\gamma}{\operatorname{argmin}} (\| y_c - X_c\gamma_c \|^2 + \lambda\|\gamma\|^2)$ ($\because d$ does not depend on γ)

$= \gamma^R(\lambda)$: ridge estimator with $k = \lambda$)

(2) $\|\hat{\gamma}(\lambda^*)\|^2 = d \Rightarrow \|\hat{\gamma}(\lambda^*)\|^2 = \|(X_c^T X_c + \lambda^* I_p)^{-1} X_c^T y_c\|^2$

$$= \|(I_p + \lambda^*(X_c^T X_c)^{-1})^{-1}(X_c^T X_c)^{-1} X_c^T y_c\|^2$$

$$= \|(I_p + \lambda^*(X_c^T X_c)^{-1})^{-1}\hat{\gamma}\|^2$$

$$= \hat{\gamma}^T (I_p + \lambda^*(X_c^T X_c)^{-1})^{-2}\hat{\gamma}$$

$$\therefore d = \hat{\gamma}^T (I_p + \lambda^*(X_c^T X_c)^{-1})^{-2}\hat{\gamma}$$

However, we also know that $d = \hat{\gamma}^T (I_p + k(X_c^T X_c)^{-1})^{-2}\hat{\gamma}$

$$\therefore \lambda^* = k$$

$$\therefore \hat{\gamma}(\lambda^*) = \hat{\gamma}^R(\lambda^*) = \hat{\gamma}^R(k)$$

(2nd equation)

$$\|y_c - X_c\gamma\|^2 = \|y_c - X_c\hat{\gamma} + X_c\hat{\gamma} - X_c\gamma\|^2$$

$$= \|y_c - X_c\hat{\gamma}\|^2 + \|X_c(\hat{\gamma} - \gamma)\|^2 - 2(\gamma - \hat{\gamma})^T \underbrace{X_c^T(y_c - X_c\hat{\gamma})}_{X_c^T(y_c - \Pi_{X_c} y_c)\, =\, X_c^T(I_p - \Pi_{X_c})y_c = 0}$$

$$= \|y_c - X_c\hat{\gamma}\|^2 + \|X_c(\hat{\gamma} - \gamma)\|^2$$

$$\therefore \underset{\gamma}{\operatorname{argmin}} \|y - X_c\gamma\|^2 = \underset{\gamma}{\operatorname{argmin}} \|y - X_c\gamma\|^2 + \|X_c(\hat{\gamma} - \gamma)\|^2$$

$$= \underset{\gamma}{\operatorname{argmin}} \|X_c(\hat{\gamma} - \gamma)^2\|$$

$$= \underset{\|\gamma\|^2 \le d}{\operatorname{argmin}} (\gamma - \hat{\gamma})^T X_c^T X_c (\gamma - \hat{\gamma})$$

Q15) Prove the following remark:

$$\|\hat{\gamma}\|^2 > d$$

$$\hat{\gamma} = \underset{\gamma \in \mathbb{R}^p}{\operatorname{argmin}} \|y_c - X_c\gamma\|^2$$

if $\|\hat{\gamma}\|^2 \le d$, $\hat{\gamma} = \underset{\gamma \in \mathbb{R}^p : \|\gamma\|^2 \le d}{\operatorname{argmin}} \|y_c - X_c\gamma\|^2$

* $\|y_c - X_c\gamma\|^2$를 최소화하는 $\gamma(=\hat{\gamma})$이 constraint 구간 안에 있으면 constraint의 의미가 없어진다.

또한 $\|\hat{\gamma}\| \le d$ 이면 $\hat{\gamma}^R(k) = \hat{\gamma}$ 을 뜻한다. 즉,

$$\hat{\gamma}^R(k) = (X_c^T X_c + kI_p)^{-1} X_c^T y = (X_c^T X_c)^{-1} X_c^T y = \hat{\gamma} \qquad \therefore k = 0$$

하지만 처음 가정에서 $k \ne 0$ 이기 때문에 모순이 발생하게 된다.
(↑ proof by contradiction)

다음으로는 수식으로 증명해 보도록 하자.

$$d = \hat{\gamma}^T (I_p + k(X_c^T X_c)^{-1})^{-2} \hat{\gamma}$$
$$= \hat{\gamma}^T (I_p + k(P\Lambda P^T)^{-1})^{-2} \hat{\gamma}$$
$$= \hat{\gamma}^T (I_p + kP\Lambda^{-1}P^T)^{-2} \hat{\gamma}$$
$$= \hat{\gamma}^T (PP^T + P(k\Lambda^{-1})P^T)^{-2} \hat{\gamma}$$
$$= \hat{\gamma}^T (P(I_p + k\Lambda^{-1})P^T)^{-2} \hat{\gamma}$$
$$= \hat{\gamma}^T P(I_p + k\Lambda^{-1})^{-2} P^T \hat{\gamma}$$

Let $\hat{\delta} = P^T\hat{\gamma} = [\hat{\delta}_1 \cdots \hat{\delta}_p]^T$

$$\hat{\gamma}^T P(I_p + k\Lambda^{-1})^{-2} P^T \hat{\gamma} = \hat{\delta}^T (I_p + k\Lambda^{-1})^{-2} \hat{\delta}$$

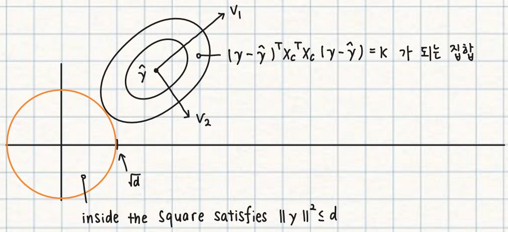
$$= \sum_{i=1}^{p} \left(1 + k\frac{1}{\lambda_i}\right)^{-2} \delta_i^2$$

$$= \sum_{i=1}^{p} \left(\frac{\lambda_i}{k + \lambda_i}\right)^2 \delta_i^2 < \sum_{i=1}^{p} \delta_i^2 = \|\hat{\delta}\|^2 = \|P^T\hat{\gamma}\|^2 = \|\hat{\gamma}\|^2$$

$$\therefore d < \|\hat{\gamma}\|^2$$

(Geometric Interpretation)

The penalty term $k\|\gamma\|^2$ of the least square criterion makes the OLS $\hat{\gamma}$ shrink towards 0.



$(\gamma - \hat{\gamma})^T X_c^T X_c (\gamma - \hat{\gamma}) = k$: 중심이 $\hat{\gamma}$이며 $X_c^T X_c$에 의해 모양이 결정되는 타원 모형이 형성된다.

v_2, v_1은 $X_c^T X_c$의 eigenvector이며 축의 길이는 λ_1, λ_2가 결정한다.

* The sketch of proof for Theorem 10 can be broken down to two fundamental ideas.

① Prove that $\underset{\gamma \in \mathbb{R}^p : \|\gamma\|^2 \le d}{\operatorname{argmin}} \|y_c - X_c\gamma\|^2$ is equivalent to $\operatorname{argmin} L(\gamma, \lambda)$ (i.e., Lagrange Multiplier) and the $L(\gamma, \lambda)$ is equivalent a ridge regression equation.

② Using the two independent equations expressing d, show that $\lambda^* = k$ and thus $\hat{\gamma}^R(\lambda^*) = \hat{\gamma}^R(k)$

Q16) Prove Theorem 11

Consider the following MLRM

$$y_c = X_c\gamma + \varepsilon,\ \varepsilon_1,\cdots,\varepsilon_n \overset{iid}{\sim} N(0,\sigma^2)$$

$$E[\hat{\gamma}^R(k)] = \gamma - k(X_c^TX_c + kI_p)^{-1}\gamma$$

$$Var(\hat{\gamma}^R(k)) = \sigma^2(X_c^TX_c + kI_p)^{-1}(X_c^TX_c)(X_c^TX_c + kI_p)^{-1}$$

$$MSE(\hat{\gamma}^R(k)) = \sum_{j=1}^{p} \frac{\lambda_j\sigma^2 + k^2\delta_j^2}{(\lambda_j+k)^2},\ \delta = P^T\gamma$$

$$\hat{\gamma}^R(k) = (X_c^TX_c + kI_p)^{-1}X_c^Ty$$

$$\begin{aligned}
E[\hat{\gamma}^R(k)] = E[(X_c^TX_c + kI_p)^{-1}X_c^Ty] &= (X_c^TX_c + kI_p)^{-1}X_c^TX_c\gamma \\
&= (X_c^TX_c + kI_p)^{-1}(X_c^TX_c + kI_p - kI_p)\gamma \\
&= \gamma - k(X^TX + kI_p)^{-1}\gamma
\end{aligned}$$

$$\begin{aligned}
Var(\hat{\gamma}^R(k)) &= (X_c^TX_c + kI_p)^{-1}X_c^T Var(y) X_c (X_c^TX_c + kI_p)^{-1} \\
&= \sigma^2(X_c^TX_c + kI_p)^{-1}X_c^TX_c(X_c^TX_c + kI_p)^{-1}
\end{aligned}$$

$$MSE(\hat{\gamma}^R(k)) = E[(\hat{\gamma}^R - \gamma)^2]$$

$$* E[y^TAy] = trace(A\Sigma) + \mu^TA\mu,\ \mu = E[y]$$

let $y = (\hat{\gamma}^R - \gamma)$, $A = I_n$ and $\Sigma = Var(y)$

$$\therefore E[(\hat{\gamma}^R - \gamma)^2] = Var(\hat{\gamma}^R - \gamma) + E[\hat{\gamma}^R - \gamma]^T E[\hat{\gamma}^R - \gamma]$$

$$\begin{aligned}
&Var(\hat{\gamma}^R) \\
&= \sigma^2(X_c^TX_c + kI_p)^{-1}X_c^TX_c(X_c^TX_c + kI_p)^{-1} \\
&= \sigma^2(P\Lambda P^T + kI_p)^{-1}(P\Lambda P^T)(P\Lambda P^T + kI_p)^{-1} \\
&= \sigma^2 P(\Lambda + kI_p)^{-1}P^TP\Lambda P^TP(\Lambda + kI_p)^{-1}P^T \\
&= \sigma^2[v_1 \cdots v_p]^T diag(\lambda_i/(\lambda_i+k)^2)[v_1 \cdots v_p] \\
&= \sigma^2\sum_{i=1}^{p}\frac{\lambda_i}{(\lambda_i+k)^2}v_i^Tv_i = \sum_{j=1}^{p}\frac{\lambda_i\sigma^2}{(\lambda_i+k)^2} \cdots (1)
\end{aligned}$$

$$\begin{aligned}
&E[\hat{\gamma}^R - \gamma]^T E[\hat{\gamma}^R - \gamma] \\
&= [k(X_c^TX_c + kI_p)^{-1}\gamma]^T[k(X_c^TX_c + kI_p)^{-1}\gamma] \\
&= k^2\gamma^T(X_c^TX_c + kI_p)^{-2}\gamma \\
&= k^2\gamma^T(P\Lambda P^T + kI_p)^{-2}\gamma \\
&= k^2\delta^T(\Lambda + kI_p)^{-2}\delta,\ \delta = P^T\gamma \\
&= k^2\sum_{i=1}^{p}\frac{\delta_i^2}{(\lambda_i+k)^2} \cdots (2)
\end{aligned}$$

$$(1)+(2) = \sum_{i=1}^{p}\frac{\sigma^2\lambda_i + k^2\delta_i^2}{(\lambda_i+k)^2},\ P^T\lambda = \delta = [\delta_1,\cdots,\delta_p]^T$$

Q17) Prove Theorem 12

Consider the MLRM of centered version:

$$y_c = X_c\gamma + \varepsilon,\ \varepsilon_1,\cdots,\varepsilon_n \overset{iid}{\sim} (0,\sigma^2)$$

Let $\hat{\gamma}^R(k)$ denote the ridge estimation for some $k \geq 0$. In this case, the map

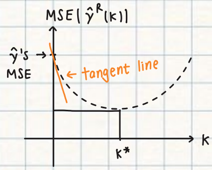
$$k \mapsto MSE[\hat{\gamma}^R(k)] : [0,\infty) \longrightarrow \mathbb{R}$$

is decreasing around zero

$$MSE[\hat{\gamma}^R(k)] = \sum_{i=1}^{p}\frac{\lambda_i\sigma^2 + k^2\delta_i^2}{(\lambda_i+k)^2}$$

$$\frac{\partial MSE[\hat{\gamma}^R(k)]}{\partial k} = \sum_{i=1}^{p} -\frac{2\lambda_i\sigma^2}{(\lambda_i+k)^3} + \frac{2k\delta_i^2}{(\lambda_i+k)^2} - \frac{2k^2\delta_i^2}{(\lambda_i+k)^3}$$

$$\left.\frac{\partial MSE[\hat{\gamma}^R(k)]}{\partial k}\right|_{k=0} = \sum_{i=1}^{p} -2\frac{\sigma^2}{\lambda_i^2} < 0$$



$$\therefore MSE(\hat{\gamma}) \geq MSE(\hat{\gamma}^R(k))$$

Although the ridge estimator is biased, it has a lower MSE than an unbiased estimator.

If multicollinearity exists, there exists an eigenvalue $\lambda_i \approx 0$. Therefore the gradient of the MSE is: $-\sum_{j=1}^{p} 2\sigma^2/\lambda_j^2 \approx -\infty$. Meaning, the MSE of the estimators is reduced rapidly for a unit increase of k.

PRINCIPLE COMPONENT REGRESSION.

$$\Lambda = \begin{bmatrix} \Lambda_1 & 0 \\ 0 & \Lambda_2 \end{bmatrix}, \quad \Lambda_1 = diag(\lambda_1, \cdots, \lambda_q), \quad \Lambda_2 = diag(\lambda_{q+1}, \cdots, \lambda_p),$$

$$P = [P_1 \; P_2], \quad P_1 = [v_1 \cdots v_q], \quad P_2 = [v_{q+1} \cdots v_p]$$
$$Z = [z_1, \cdots, z_p] = X_c P$$

Q18) Show that Z may be partitioned so that

$$Z_1 = [z_1, \cdots, z_q] = X_c P_1$$
$$Z_2 = [z_{q+1}, \cdots, z_p] = X_c P_2$$

$$\begin{aligned} X_c P &= X_c [P_1 \; P_2] \\ &= X_c [v_1 \cdots v_q \; v_{q+1} \cdots v_p] \\ &= [X_c v_1 \cdots X_c v_q \; X_c v_{q+1} \cdots X_c v_p] \\ &= [z_1, \cdots, z_q, z_{q+1}, \cdots, z_p] \end{aligned}$$

(proof)

$$P^T P = I_p \Rightarrow \underbrace{\begin{bmatrix} P_1^T \\ P_2^T \end{bmatrix} [P_1 \; P_2]} = \underbrace{[P_1 \; P_2] \begin{bmatrix} P_1^T \\ P_2^T \end{bmatrix}}_{P_1 P_1^T + P_2 P_2^T = I_p} = I_p$$

$$\begin{bmatrix} P_1^T P_1 & P_1^T P_2 \\ P_2^T P_1 & P_2^T P_2 \end{bmatrix} = \begin{bmatrix} I_q & 0 \\ 0 & I_{p-q} \end{bmatrix}$$

$\therefore$ we derive 3 equations:

(1) $P_1^T P_1 = I_q$
(2) $P_2^T P_2 = I_{p-q}$
(3) $PP_1^T + P_2 P_2^T = I_p$

$$\Lambda_1 = diag(\cdots \lambda_1, \cdots, \lambda_q)$$
$$\Lambda_2 = diag(\lambda_{q+1}, \cdots, \Lambda_p)$$
$$Z = X_c P = (z_1, \cdots, z_p)$$

⊛ $\|z_j\|^2 = \lambda_j$ & $\langle z_j, z_i \rangle = 0 \quad (i \neq j)$

we derived this from the fact that $Z^T Z = \Lambda$

$$Z^T Z = P^T X_c^T X_c P = P^T P \Lambda P^T P = \Lambda$$

$$\Rightarrow \begin{pmatrix} Z_1^T \\ Z_2^T \end{pmatrix} (Z_1 \; Z_2) = \Lambda = \begin{pmatrix} \Lambda_1 & 0 \\ 0 & \Lambda_2 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} Z_1^T Z_1 & 0 \\ 0 & Z_2^T Z_2 \end{pmatrix} = \begin{pmatrix} \Lambda_1 & 0 \\ 0 & \Lambda_2 \end{pmatrix}$$

$\therefore Z_1^T Z_1 = \Lambda_1$ and $Z_2^T Z_2 = \Lambda_2$

Q19) Consider the MLRM of centered version. Represent the model by Principle Components.

$$y_c = X_c \gamma + \epsilon, \quad \epsilon_1, \cdots, \epsilon_n \overset{iid}{\sim} (0, \sigma^2)$$

$$\begin{aligned} y_c &= X_c P P^T \gamma + \epsilon \\ &= Z\alpha + \epsilon \quad (Z = X_c P, \; \alpha = P^T \gamma) \\ &= Z_1 \alpha_1 + Z_2 \alpha_2 + \epsilon \quad (\alpha_1 = P_1^T \gamma \text{ and } \alpha_2 = P_2^T \gamma) \end{aligned}$$

In the case of multicollinearity, where $\Lambda_2 \approx 0$, we have shown

$$Z_2 \approx (Z_{q+1}, \cdots, Z_p) \approx 0 \quad (\because \lambda_j \approx 0 \Leftrightarrow Z_j \approx 0)$$

Therefore, our model is approximately equal to the reduced model:

$$\begin{aligned} y_c &= Z\alpha + \epsilon \quad \text{with } \alpha_2 = 0 \\ &= Z_1 \alpha_1 + \epsilon \quad (\because Z\alpha = Z_1 \alpha_1 + Z_2 \alpha_2 \approx Z_1 \alpha_1) \end{aligned}$$

Q20) What is the Least Square estimator for the reduced model ($y_c = Z_1 \alpha_1 + \epsilon$)?

$$\hat{\alpha}_1 = (Z_1^T Z_1)^{-1} Z_1^T y_c \quad \text{and} \quad \hat{\alpha} = (\hat{\alpha}_1^T, 0^T)^T$$

Q21) Define the PCR estimator for γ

$$\hat{\gamma}^{PCR} = P\hat{\alpha}, \quad \hat{\alpha} = (\hat{\alpha}_1^T \; 0^T)^T$$

Q22) Prove Theorem 14

Consider the centered MLRM and let $\hat{\gamma}^{PCR}$ denote the PCR estimator for some $q > p$. In this case, show the following:

$$E[\hat{\gamma}^{PCR}] = \gamma - P_2 P_2^T \gamma$$
$$Var(\hat{\gamma}^{PCR}) = \sigma^2 P_1 \Lambda_1^{-1} P_1^T$$
$$MSE(\hat{\gamma}^{PCR}) = \gamma^T P_2 P_2^T \gamma + \sum_{i=1}^{q} \frac{\sigma^2}{\lambda_i}$$

$$\begin{aligned} E[\hat{\gamma}^{PCR}] = E[P\hat{\alpha}] &= P_1 E[(Z_1^T Z_1)^{-1} Z_1^T y_c] = P_1 [(P_1^T X_c^T X_c P_1)^{-1} (X_c P_1)^T y_c] \\ &= P_1 (P_1^T (X_c^T X_c)^{-1} P_1 P_1^T X_c^T X_c \gamma) \\ &= P_1 P_1^T \gamma \\ &= (I - P_2 P_2^T) \gamma \\ &= \gamma - P_2 P_2^T \gamma \end{aligned}$$

$$\begin{aligned} Var(\hat{\gamma}^{PCR}) = Var(P_1 \hat{\alpha}_1) &= P_1 Var(\hat{\alpha}_1) P_1^T = P_1 Var((Z_1^T Z_1)^{-1} Z_1^T y_c) P_1 \\ &= P_1 (Z_1^T Z_1)^{-1} Z_1^T Var(y_c) Z_1 (Z_1^T Z_1)^{-1} P_1 \\ &= \sigma^2 P_1 (Z_1^T Z_1)^{-1} P_1 \\ &= \sigma^2 P_1 \Lambda^{-1} P_1 \end{aligned}$$

$$MSE[\hat{y}^{PCR}] = y^T P_2 P_2^T y + \sum_{j=1}^{q} \frac{\sigma^2}{\lambda_j}$$

$$MSE[\hat{y}^{PCR}] = E[(\hat{y}^{PCR} - y)^2] = E[y^T A y] = trace(A\Sigma) + \mu^T A \mu$$

$* \; y = (\hat{y}^{PCR} - y) \quad A = I_p, \; \Sigma = Var(y), \; \mu = E[y]$

$$= \underbrace{trace(\sigma^2 P_1 \Lambda_1^{-1} P_1^T)}_{(1)} + \underbrace{(y - P_2 P_2^T y)^T (y - P_2 P_2^T y)}_{(2)}$$

(1) $trace(\sigma^2 P_1 \Lambda_1^{-1} P_1^T) = trace(\sigma^2 \Lambda_1^{-1} P_1^T P_1) = \sigma^2 trace(\Lambda_1^{-1}) = \sum_{j=1}^{q} \frac{\sigma^2}{\lambda_j}$

(2) $E[(\hat{y}^{PCR} - y)] = P_2 P_2^T y_2$

$\therefore E[(\hat{y}^{PCR} - y)]^T E[(\hat{y}^{PCR} - y)] = y^T P_2 P_2^T P_2 P_2^T y = y^T P_2 P_2^T y$

$\therefore MSE[\hat{y}^{PCR}] = \sum_{j=1}^{q} \frac{\sigma^2}{\lambda_j} + y^T P_2 P_2^T y$

Q23) Prove Theorem 15.

Consider the centered MLRM and let $\hat{y}^{PCR}$ denote the PCR estimator for som $q<p$. Show that if the model has strong multi Collinearity, we get to know the following:

$$MSE(\hat{y}^{PCR}) < MSE(\hat{y}), \quad \hat{y} : \text{OLS estimator.}$$

$MSE(\hat{y}^{PCR}) = \sum_{j=1}^{q} \frac{\sigma^2}{\lambda_j} + y^T P_2 P_2^T y$

$MSE(\hat{y}) = \sum_{j=1}^{p} \frac{\sigma^2}{\lambda_j}$

if multicollinearity exists, eigenvalues, $\lambda_{q+1}, \cdots, \lambda_p \approx 0$

$\Leftrightarrow \sum_{j=q+1}^{p} \frac{\sigma^2}{\lambda_j} \gg 0 \Rightarrow$ increase in MSE.

* $\hat{y}^{PCR}$은 biased estimator 이지만, 분산(variance)가 $\hat{y}$보다 낮게 나온다는 장점이 있다.

* MSE는 bias와 variance, 이 두 가지 측정기준을 복합적으로 고려하는 기준이다.

* Multicollinearity가 있을 때 $MSE(\hat{y}) > MSE(\hat{y}^{PCR})$이 더 두드러지게 드러난다.